📘 Hously Finttech Realty — Documentation Part 3

Test Cases, Security & Deployment

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11. Test Cases — Industry Standard Format

> **Format:** Each test case follows the IEEE 829 standard with Test ID, Title, Module, Priority, Precondition, Steps, Expected Result, and Status.

11.1 Module: Authentication (TC-AUTH)

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-AUTH-001	Valid Login	🔴 Critical	Server running, user exists in DB	1. Click navbar right button 2. Click sidebar logo 3. Enter `kamleshshah@gmail.com` / `Kamlesh@123` 4. Click Login	200 OK, httpOnly cookies set, user data returned, sidebar closes	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-AUTH-002	Invalid Email Format	🟠 High	Login modal open	1. Enter `notanemail` 2. Click Login	422 Validation Error: "Enter a valid email address"	✅ Pass
TC-AUTH-003	Wrong Password	🟠 High	Valid user exists	1. Enter correct email 2. Enter wrong password 3. Click Login	401 Unauthorized: "Invalid credentials"	✅ Pass
TC-AUTH-004	Empty Fields	🟡 Medium	Login modal open	1. Leave email/password blank 2. Click Login	422 Validation Error for each required field	✅ Pass
TC-AUTH-005	Token Auto-Refresh	🔴 Critical	User logged in, accessToken near expiry	1. Wait 15 min for accessToken to expire 2. Make any API call	Axios interceptor silently calls `/api/auth/refresh`, retries request, user stays logged in	✅ Pass
TC-AUTH-006	Refresh Token Expired	🟠 High	Both tokens expired	1. Let both tokens expire (7 days) 2. Make API call	Redirected to `/`, sessionStorage cleared	✅ Pass
TC-AUTH-007	Logout Single Device	🟡 Medium	User logged in	1. Click Logout in sidebar	200 OK, cookies cleared, user returned to home	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-AUTH-008	Logout All Devices	🟡 Medium	User logged in	1. Call `POST /api/auth/logout-all`	All refresh tokens for user deleted from DB, cookies cleared	✅ Pass
TC-AUTH-009	Protected Route Without Auth	🔴 Critical	No cookies set	1. Directly access `/api/auth/me`	401: "Access denied. Please log in."	✅ Pass
TC-AUTH-010	Create Developer Account	🟡 Medium	Admin logged in	1. Navigate to Developer Mgmt tab 2. Enter name, email, strong password 3. Submit	201 Created, new user appears in list	✅ Pass

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11.2 Module: Job Applications (TC-JOB-APP)

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-APP-001	Submit Valid Application	🔴 Critical	`/career/job-detail` page open	1. Fill all required fields 2. Upload PDF resume 3. Click Submit	201 Created, success message shown, record in DB	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-APP-002	Submit Without Resume	🟠 High	Apply form open	1. Fill all fields 2. Skip resume upload 3. Submit	Resume stored as NULL, application saved	✅ Pass
TC-JOB-APP-003	Submit With Invalid File Type	🟡 Medium	Apply form open	1. Upload `.exe` or `.jpg` as resume 2. Submit	Multer accepts file (MIME stored), or validation error if restricted	⚠️ Check
TC-JOB-APP-004	View All Applications (Admin)	🔴 Critical	Admin logged in	1. Dashboard → Job Applications tab	Table loads with all applications, no error	✅ Pass
TC-JOB-APP-005	View Resume PDF	🟠 High	Application with resume exists	1. Click “View Resume” button	Browser opens PDF at `/api/jobs/resume/:id` in new tab	✅ Pass
TC-JOB-APP-006	Update Application Status	🟠 High	Admin logged in, application exists	1. Select status dropdown → "selected" 2. Save	200 OK, status updated in DB and reflected in UI	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-APP-007	Delete Application	🟡 Medium	Admin logged in	1. Click Delete on application row 2. Confirm	Application removed from table and DB	✅ Pass
TC-JOB-APP-008	Unauthenticated Access to Applications	🔴 Critical	No auth cookies	1. `GET /api/jobs/` without login	401: “Access denied. Please log in.”	✅ Pass

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11.3 Module: Job Opportunities / Postings (TC-JOB-POST)

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-POST-001	Create New Job Posting	🔴 Critical	Admin logged in	1. Dashboard → Job Postings 2. Click "Add Job" 3. Fill all fields 4. Submit	201 Created, job appears in list and on `/career/job` page	✅ Pass
TC-JOB-POST-002	Create Job Missing Title	🟠 High	Admin logged in	1. Create job without title	Validation error or DB constraint failure	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-POST-003	JSON Array Fields (Responsibilities)	🟡 Medium	Create job form	1. Enter multiple responsibilities 2. Submit	Stored as JSON array in DB, parsed correctly on fetch	✅ Pass
TC-JOB-POST-004	Public Job Listing Fetch	🔴 Critical	Jobs exist in DB	1. Navigate to `/career/job` (no login)	All jobs displayed, no auth required	✅ Pass
TC-JOB-POST-005	Update Job Posting	🟠 High	Admin logged in, job exists	1. Click Edit on job 2. Change salary 3. Save	200 OK, updated data shown in list	✅ Pass
TC-JOB-POST-006	Delete Job Posting	🟡 Medium	Admin logged in	1. Click Delete on job	Job removed from list and DB	✅ Pass
TC-JOB-POST-007	Job Search & Filter	🟡 Medium	`/career/job` open, jobs exist	1. Type keyword in search 2. Select type filter 3. Select location filter	Jobs filtered correctly in real-time	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-JOB-POST-008	isNew Badge Display	🟢 Low	Job with `isNew=true` exists	1. View job listing	“NEW” badge shown on job card	✅ Pass

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11.4 Module: Project Requests (TC-PROJ)

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-PROJ-001	Submit Valid Project Inquiry	🔴 Critical	`/contact` page open	1. Fill name, email, inquiry type, service, message 2. Submit	201 Created, success message shown	✅ Pass
TC-PROJ-002	Submit Without Required Fields	🟠 High	Contact form open	1. Leave email blank 2. Submit	Validation error displayed	✅ Pass
TC-PROJ-003	View All Project Requests (Admin)	🔴 Critical	Admin logged in	1. Dashboard → Project Requests tab	All inquiries listed in table	✅ Pass
TC-PROJ-	Update Project Status	🟡 Medium	Admin, request exists	1. Change status to "reviewed"	Status updated in DB and UI	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
004				2. Save		
TC-PROJ-005	Delete Project Request	🟡 Medium	Admin logged in	1. Click Delete on request	Removed from list and DB	✅ Pass

11.5 Module: Developer Management (TC-DEV)

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-DEV-001	Create Developer - Valid	🔴 High	Admin logged in	1. Fill name, email, strong password (8+ chars, uppercase, number, special char) 2. Submit	201 Created, developer added to user list	✅ Pass
TC-DEV-002	Create Developer - Weak Password	🟡 Medium	Admin logged in	1. Enter password: `weak`	422 Validation: password requirements not met	✅ Pass
TC-DEV-003	Create Developer - Duplicate Email	🟡 Medium	User with email exists	1. Enter existing email	DB unique constraint error, 409 or 500 with message	✅ Pass

TC ID	Test Case	Priority	Precondition	Test Steps	Expected Result	Status
TC-DEV-004	Delete Developer	🟡 Medium	Developer user exists	1. Click Delete on user row	User removed from list and DB	✅ Pass
TC-DEV-005	List All Users	🟢 Low	Multiple users in DB	1. Dashboard → Developer Mgmt	All users listed in table with id, name, email, date	✅ Pass

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11.6 Module: Frontend Pages (TC-UI)

TC ID	Test Case	Priority	Test Steps	Expected Result
TC-UI-001	Home Page Load	🔴 Critical	Navigate to `/`	Hero section, services, testimonials render without error
TC-UI-002	About Page	🟡 Medium	Navigate to `/about`	About content loads, no broken images
TC-UI-003	Services Page	🟡 Medium	Navigate to `/services`	All service cards displayed
TC-UI-004	Contact Form Submission	🟠 High	`/contact` → fill form → submit	Success toast shown, data saved

TC ID	Test Case	Priority	Test Steps	Expected Result
TC-UI-005	Career Page	🟡 Medium	Navigate to `/career`	Career content displayed, CTA links working
TC-UI-006	Job Listings Page	🟠 High	Navigate to `/career/job`	Jobs fetched and displayed, filters work
TC-UI-007	Job Detail + Apply	🔴 Critical	Click job → fill form → submit	Application submitted successfully
TC-UI-008	Responsive Design	🟡 Medium	Resize to 375px (mobile)	Navbar collapses, layout remains usable
TC-UI-009	Navigation Links	🟡 Medium	Click each nav link	Correct route loaded, no 404s
TC-UI-010	Swiper Testimonials	🟢 Low	Scroll to testimonials	Carousel auto-plays, manual swipe works

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12. Input Validation Rules

12.1 Registration / Create Developer

Field	Rule
`name`	Required, 2–60 characters
`email`	Required, valid email format, normalized
`password`	Required, min 8 chars, 1 uppercase, 1 number, 1 special char

12.2 Login

Field	Rule
`email`	Required, valid email format
`password`	Required (non-empty)

12.3 Job Application

Field	Rule
`full_name`	Required
`email`	Required, valid email
`phone`	Required
`experience`	Required

Field	Rule
`skills`	Required
`tenth_percentage`	Numeric 0–100
`twelfth_percentage`	Numeric 0–100
`graduation_percentage`	Numeric 0–100
`resume`	File upload (stored as BLOB)

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13. Security Checklist

Security Control	Implementation	Status
Password Hashing	`bcrypt` with salt rounds	✅
JWT in httpOnly Cookies	Not accessible via JS, XSS-safe	✅
CORS Configured	Only `CLIENT_URL` origin allowed	✅
Token Expiry	Access: 15min, Refresh: 7 days	✅
Input Validation	`express-validator` on all inputs	✅
SQL Injection Prevention	Parameterized queries (`mysql2`)	✅

Security Control	Implementation	Status
Protected Routes	`protect` middleware on all admin endpoints	✅
Refresh Token Rotation	Old token deleted on each refresh	✅
Error Messages	Stack trace hidden in production	✅
File Upload Security	`multer` memory storage, MIME type stored	⚠️ Add type restrictions

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14. Environment Variables

```
# Server/.env

# Database
DB_HOST=localhost
DB_USER=root
DB_PASSWORD=your_db_password
DB_NAME=hously_db
DB_PORT=3306

# JWT Secrets
JWT_SECRET=your_jwt_access_secret_key
JWT_REFRESH_SECRET=your_jwt_refresh_secret_key

# Server
PORT=5000
NODE_ENV=development
```

```
# Client
CLIENT_URL=http://localhost:5173
```

```
# Client/.env

VITE_API_URL=http://localhost:5000
```

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15. Deployment Guide

15.1 Local Development

```
# Terminal 1 — Backend
cd server
npm install
npm run dev        # nodemon server.js — hot reload on :5000

# Terminal 2 — Frontend
cd client
npm install
npm run dev        # Vite dev server on :5173
```

15.2 Production Build

```
# Frontend
cd client
npm run build      # Outputs to dist/

# Backend
cd server
npm start          # node server.js (no nodemon)
```

15.3 Database Setup

```
-- Run these in MySQL:
CREATE DATABASE hously_db;
USE hously_db;

-- Then run table creation scripts from Section 5
```

15.4 Pre-Deployment Checklist

Check	Description
✅ `.env` configured	All secrets set, `NODE_ENV=production`
✅ `CLIENT_URL` set	Points to production frontend URL
✅ `VITE_API_URL` set	Points to production backend URL
✅ MySQL DB running	Tables created, admin user seeded

Check	Description
✅ CORS origin updated	Production domain allowed
✅ HTTPS enabled	Cookies with `secure: true` in production

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16. Error Response Format

All API errors follow a consistent format:

```
{
  "success": false,
  "message": "Human-readable error message",
  "errors": [                    // Optional - validation errors
    {
      "field": "email",
      "message": "Enter a valid email address"
    }
  ],
  "stack": "..."                 // Only in development mode
}
```

Common HTTP Status Codes

Code	Meaning	When Used
`200`	OK	Successful GET/PUT/DELETE
`201`	Created	Successful POST (new record)
`401`	Unauthorized	Missing/invalid/expired token
`404`	Not Found	Resource doesn't exist
`409`	Conflict	Duplicate email (unique constraint)
`422`	Unprocessable Entity	Validation failure
`500`	Internal Server Error	DB error, unexpected failure

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17. Known Issues & Notes

Issue	Description	Priority	Fix
Resume file type restriction	No MIME type validation on upload	🟡 Medium	Add multer `fileFilter` for PDF/DOC only
Dashboard auth check	Dashboard page checks `sessionStorage` only, no server validation on mount	🔴 High	Add `GET /api/auth/me` check on Dashboard load

Issue	Description	Priority	Fix
Duplicate email error	Returns 500 instead of 409 on duplicate email	🟡 Medium	Catch `ER_DUP_ENTRY` MySQL error, return 409
`isNew` flag manual	Not auto-reset after a period	🟢 Low	Add CRON job or auto-expire after 7 days

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18. Glossary

Term	Definition
httpOnly Cookie	A browser cookie inaccessible to JavaScript; prevents XSS token theft
JWT	JSON Web Token — compact, signed token carrying user claims
Access Token	Short-lived JWT (15 min) used to authenticate API requests
Refresh Token	Long-lived token (7 days) stored in DB, used to generate new access tokens
BLOB	Binary Large Object — MySQL data type used to store resume files
CORS	Cross-Origin Resource Sharing — controls which domains can access the API
Axios Interceptor	Code that runs before/after every Axios request; used for auto token refresh
protect middleware	Express middleware that validates the accessToken cookie

Term	Definition
Repository Layer	Data access layer — all SQL queries isolated here
Service Layer	Business logic layer between controllers and repositories

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End of Hously Finttech Realty Technical Documentation — v1.0.0

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